Birla Institute of Technology and Science, Pilani
Second Semester: 2025-2026
DRM Project: Pricing Futures and Margin Account Management
(*12% Weightage*)
Deadline: 6th March 2026

Project Guidelines

This project is divided into three sections, and all parts must be completed for evaluation. Each group will be assigned two companies: one large-cap and one small-cap. Please clearly list all assumptions and data used in either the Excel file or the analysis document. After submission, a presentation will be scheduled after the mid-semester exam.

Detailed Task Description

Section A: Data Collection & Statistics (2%)

1. Utilise financial data APIs (such as yfinance) to retrieve daily stock price data for the specified companies. The time frame must be from March 4, 2025, to March 4, 2026. Focus solely on closing price data for any subsequent analysis.

2. Using the extracted stock price data, perform a rudimentary statistical analysis on the two stocks. Then compare the two stocks using the statistical parameters you have chosen and analyse them.

Section B: Pricing Futures (5%)

1. Data Collection

- Collect January futures price data for the assigned large-cap stock (February and March expiries).
- Collect and report all additional inputs required for futures pricing, including:
    - Risk-free interest rates for relevant maturities.
    - Dividend information during the life of the futures contracts.
    - Any other costs or benefits associated with holding the underlying asset.

2. Theoretical Futures Pricing

Using the collected market data, students are required to theoretically price futures contracts **daily** on the given stock for the following maturities:

- Feb 2026 Expiry Future
- Mar 2026 Expiry Future
- Future expiring on 4th February 2027.

Each futures price must be computed separately, accounting for the maturity-specific parameters.

3. Cost-of-Carry Components

While pricing futures, students must explicitly incorporate and discuss the following components and justify excluding any component, if applicable.

- Risk-free interest rate (RBI T-Bill)
- Dividends
- Carry costs
- Convenience yield
- Transaction costs

4. Observations

Based on the computed futures prices, you should:

- Comment on the relationship between futures prices and contract maturity
- Discuss the impact of interest rates and dividends on futures pricing
- Provide relevant financial intuition for the observed results.

5. Compare with the real prices

For the Feb and Mar Fut expiries, compare the price listed on the exchange with the theoretical price you found. Calculate the difference daily. Explain this difference in prices.

Section C: Margin Call (5%)

You are managing 50 lakh rupees. You were bullish on the market forecast for January and decided to invest up to 45 lakh rupees in the large-cap futures allotted to you on 1st January 2026. Do this whole section separately for **both** the Feb Fut and Mar Fut.

The remaining cash is kept aside for margin calls, if any. If at any point this buffer falls short of covering margin call requirements, you can borrow at the overnight marginal cost of funds-based lending rate offered by SBI, plus a 2% spread. This borrowed amount will accrue interest until it is paid back at the end of the closing of all positions on 31st Jan.

Assume initial margin is 30% and maintenance margin is 20%. Each stock has a unique lot size and based on this, get as many futures contracts as possible within 45 lakhs rupees, keeping the rest as cash buffer. All positions are marked to market at the end of each trading day.

1. Compute the changes in the Margin account daily using a table.

2. Calculate the final profit or loss at the end of closure of the position, including the effect of borrowed cash (if any).

3. Assuming you used a forward contract instead of trading on futures, what difference would it make, and how much would the profit or loss be?

Deliverables

1. Excel file containing all raw data, cleaned data, and pricing calculations used in the analysis.

2. Word document including assumptions, methodology, analysis, results, and observations for all sections.

3. Python code used for data extraction, analysis, futures pricing, and margin account computation.